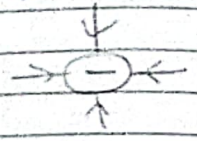
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Electricity

Charge :-

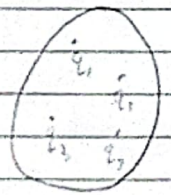
→ It is the fundamental property of matter.
→ That experience a force of attraction or repulsion.
→ when placed in an electromagnetic field.

Positive charge (+)	Negative Charge (-)
	
→ Field lines always originate from positive charge.	→ Field lines always enters into negative charge.
→ An object on losing electrons acquires positive charge.	→ An object on gaining electrones acquires negative charge.

1. **Electric Field Lines:** It is the space around a charge where another experience a force of attraction or repulsion.

Properties of charge :

1. q → individual charge
Q → Total charge
$Q = q_1 + q_2 + q_3 + q_4 + q_n$
↳ (Additivity of charge)
↳ Addition



2. $q_1 = q_2 = q_3 = q_4 = q_n$

$Q = ne$
- e → charge of an electron ($1.6 \times 10^{-19} C$)
- n → no. of charge

(Quantisation of charge)

3. **Conservation of charge :**
Charge can neither be created nor be destroyed. it can only be converted from one form to another.

- **Electric Current** - Flow of charge in a conductor for a given time interval.

OR,

Rate of flow of charge in a conductor

$I = \frac{Q}{t}$
- Q → coulomb (C)
- t → sec (s)
- I → Ampere (A)

Questions

Q1 Define one coulomb.

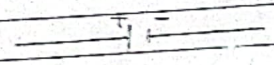
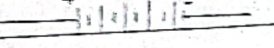
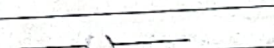
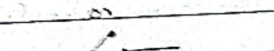
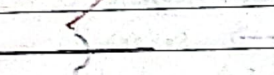
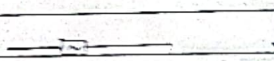
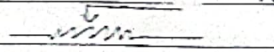
Ans $I = \frac{q}{t}$ $\quad q = I \times t$

$1C = 1A \cdot 1s$

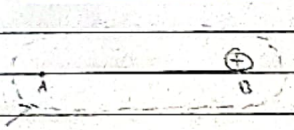
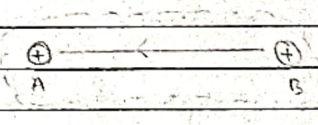
1 coulomb of charge is defined as when one ampere of current flows through a conductor in one second.

- **Electric circuit**- It is a closed path for the flow of electric current containing different electrical components.

- **Parts of an electric circuit**-

1.  Cell
2.  battery
3.  Key (open)
4.  Key (closed)
5.  Resistance
6.  wires crossing each other
7.  electrical fuse
8.  Ammeter
9.  Voltmeter
10.  Galvanometer
11.  Resistors

* Cell - combination of cathode → Anode.
* Battery - combination of cells
* Key (open) - no current will flow in circuit.
* Key (closed) - current will flow in circuit.
* Resistence - opposes the flow of current in a circuit.
* Resistor - provides variable resistance.
* Fuse - prevents an electric circuit from overheating
* Ammeter - measures the flow of current in a circuit (higher value)
- It is connected in series to a circuit.
* Voltmeter - measures potential difference across the circuit.
- It is connected in parallel to a circuit.
* Galvanometer - measures the flow of current in a circuit (smaller value).
- It also tells the direction of flow of current in a circuit.
- It is connected in series to a circuit.

Electrical Potential	Potential Difference (V)
	
$V = \frac{W}{q}$	$V_{BA} = V_B - V_A$
→ It is the amount of work done in moving a test charge from infinity to a point on the conductor in the region of its electric field.	→ It is the amount of work done in moving a test charge from one point to another on a conductor in the region of its electric field.

Ohm's Law:-

At constant temperature,

$$I \propto V$$

$$I \propto \frac{1}{R}$$

$$I \propto \frac{V}{R}$$

$$I = K \frac{V}{R}$$

$$[K = 1]$$

$$I = \frac{V}{R}$$

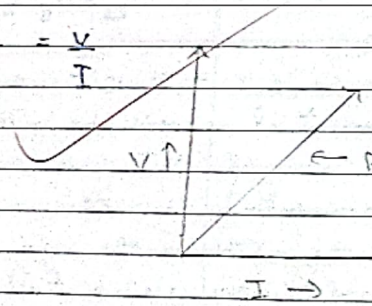
$$IR = V$$

Definition:

It states that at constant temperature, current flowing in a conductor is directly proportional to the applied potential difference and inversely proportional to the resistance of the conductor.

Graph for Ohm's Law:

$$R = \frac{y}{x} = \frac{V}{I}$$



Factors on which resistance of a conductor depends:-

1. Length of Conductor

$$R \propto l$$

2. Area of Cross Section

$$R \propto \frac{1}{A}$$

3. Material of Conductor

4. Temperature

$$R \propto T$$

$$R \propto l \qquad R = \rho \frac{l}{A}$$

$$R \propto \frac{1}{A} \qquad (\rho = \text{Resistivity})$$

$$R \propto \frac{l}{A}$$

Resistivity

→ Material resistance of a conductor.

→ S.I unit is $\frac{R \times A}{l} = \rho$

$$\frac{\Omega \cdot m^2}{m} = \Omega m$$

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Questions

Q1 Define one Ampere.

Ans One Ampere of current is defined as when the potential difference across one Ω resistance is 1 volt.

Q2 Define one volt.

Ans 1 volt is defined as when 1 ampere of current flows across a resistance of 1Ω.

Types of flow:-

Conventional flow (flow of charge)	Electric flow (flow of e⁻)
⊕ → ⊖	⊕ ← ⊖

Combination of Resistors:

- We combine resistors in two ways:-

(i) Series Combination

(ii) Parallel Combination

- Series combination is preffered when we require constant current.
- Parallel combination is preffered when we require constant voltage.

* Series Combination:

Resistors are connected one after the other.

$R_1 \neq R_2 \neq R_3$	$V = IR$
	$V_1 = IR$
I entering = I leaving	$V_3 = IR_3$
	$V = V_1 + V_2 + V_3$
$V = V_1 + V_2 + V_3$	$IR = IR_1 + IR_2 + IR_3$
	$R = (R_1 + R_2 + R_3)$
According to Ohm's Law	Equivalent Resistance

Questions

Q1 Four resistors of resistance = 2Ω, 3Ω, 5Ω, 10Ω are connected in series to a 100 V supply.

(i) Equivalent resistance

(ii) Magnitude of current

(iii) Draw a suitable Diagrame of the above

Ans (i) The total resistance $R_1 + R_2 + R_3 + R_4$

$2\Omega + 3\Omega + 5\Omega + 10\Omega = 20\Omega$

(ii) $V = IR$

$I = \frac{V}{R}$

$I = \frac{100}{20\Omega} = 4$

Electric Power (P)

It is the product of voltage and current.

SI unit = watt (w)

kilo watt (kw)

* 1 kw = 1000 w
* 1 MW = 1000 kw = 10^6 w

$$P = VI$$

Energy (E)

Product of Power and Time.

SI unit = ~~kilowatt~~ Joules

Questions

Q1 Convert 1 kilowatt hour into joules.

Solu Kilowatt = 1

hour = 1

1 kw = 1000 w = 10^3 w

1 hour = 3600 sec = 3.6×10^3 s

$\Rightarrow 10^3 \times 3.6 \times 10^3 = 3.6 \times 10^6$ Joules

---

Q2 Four electric fan of 100 watt is used for 10 hours daily, 5 electric bulb of 50 watt is used for 20 ~~watt~~ hours daily, a refrigerators of 1200 watt is used for 24 hours daily. Calcucate

(a) Total energy consumed in the mounth of June.

(b) Electricity bill at the rate of ₹8 per unit.

Solu Fan = 100 watt

Time = 10 hours daily

$P_1 = \frac{100}{1000} = 0.1 \text{ kw} \times 10 = 1 \text{ kw}$

Power of 4 fans = 4 kw

Electric bulb = 50 watt

Time = 20 hours

$P = \frac{50}{1000} = 0.05 \text{ kw} \times 20$

Power of 5 bulbs = $0.05 \times 5 = 0.25 \text{ kw} \times 20 = 5 \text{ kwhr}$

Refigerator = 1200 watt

Time = 24 hours

$P = \frac{1200}{1000} = 1.2 \text{ kw} \times 24 = 28.8 \text{ kwhr}$

Total Power consumed in one day = 28.8 + 5 + 4

= 37.8 kw

Total Power consumed in June = 37.8 × 30

= 1134 kw

Electricity bill = 1134 × 8

= 9072 rupees

Ex-11.13- Ans - - -

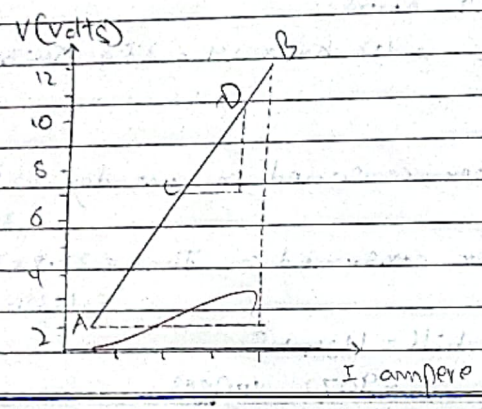
NCERT Questions

7-Solv The average value of resistance is

$\therefore R_{AB} = \frac{V}{I}$

$= \frac{13.2 - 1.6}{4 - 0.5}$

$= 3.3\,\Omega$



8-Solv Given $V = 12$ volt,

$I = 2.5$ mA

$= 2.5 \times 10^{-3}$ A

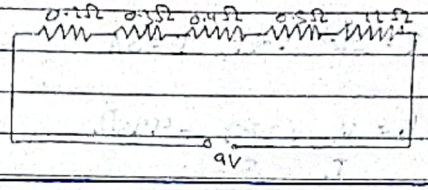
$R = \frac{V}{I} = \frac{12}{2.5 \times 10^{-3}} = 4800\,\Omega$

9-Solv Given $V = 9$ volt

Total resistance $= 0.2 + 0.3 + 0.4 + 0.5 + 12$

$= 13.4$

$I = \frac{V}{R} = \frac{9}{12.4} = 0.67$ A



10-Solv Given $V = 220$ volt,

$I = 5$A

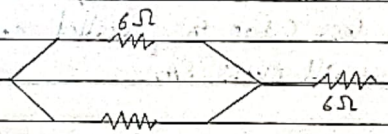
$R = \frac{V}{I} = \frac{220}{5} = 44\,\Omega$

Let n identical resistors connected in parallel.

$\frac{1}{R_{eq}} = \frac{1}{R}$

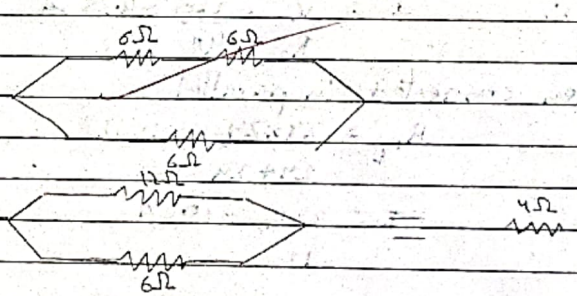
$n = \frac{R}{R_{eq}} = \frac{176}{44} = 4$

11-Solv (i)



$= 3\,\Omega + 6\,\Omega = 9\,\Omega$

(ii)



12-Solv Here $V = 220$ volt, $I = 5A$

$$\therefore \quad R = \frac{V}{I} = \frac{220}{5} = 44\,\Omega$$

i.e., a minimum resistance of $44\,\Omega$ can be applied.

Resistance of the bulb $R = \frac{V^2}{P} = \frac{220 \times 220}{10}$

$$= 4840\,\Omega$$

For parallel connection of n identical resistances, the equivalent resistance is

$$R_{eq} = \frac{R}{n}$$

$$\therefore \quad n = \frac{R}{R_{eq}} = \frac{4840}{44} = 110$$

If bulbs are more than 110, then resistance will becomes less (due to parallel combination) and current will cross 5A.

13-Solv when connected in series:

$$R_{eq} = 24 + 24 = 48\,\Omega$$

$$\therefore \quad I = \frac{V}{R_{eq}} = \frac{220}{48} = 4.58\,A$$

when connected in parallel

$$R_{eq} = \frac{24 \times 24}{24 + 24} = 12\,\Omega$$

$$I = \frac{220}{12} = 18.3\,A$$

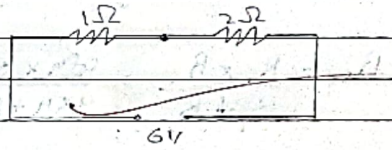
when only one is connected

$$I = \frac{220}{24} = 9.17\,A$$

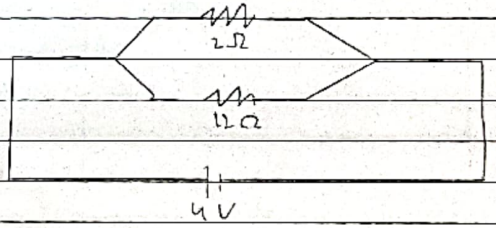
14-Solv (i) $$I = \frac{6}{1+2} = 2A$$

$\therefore$ Power used by $2\,\Omega$ resistor

$$P_1 = I^2R = (2)^2 \times 2 = 8\,W$$



(iii) $$I = \frac{4}{14} = 0.28A$$



$\therefore$ Power used by $2\,\Omega$ resistor

$$P_2 = \frac{V^2}{R} = \frac{(4)^2}{2} = 8\,W$$

$$\therefore \quad \frac{P_1}{P_2} = \frac{8}{8} \approx \frac{1}{1}$$

15-Solu Resistance of first lamp

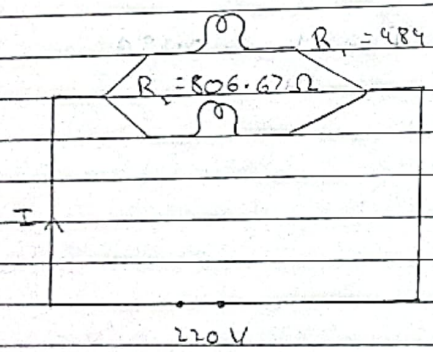
$$R_1 = \frac{V^2_{rated}}{P_{rated}} = \frac{(220)^2}{100} = 484$$

Resistance of second lamp

$$R_2 = \frac{V^2_{rated}}{P_{rated}} = \frac{(220)^2}{60} = 806.67\ \Omega$$

For parallel combination equivalent resistance

$$R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{484 \times 806.67}{484 + 806.67} = 302.5\ \Omega$$



Therefore, current drawn I is

$$I = \frac{V}{R} = \frac{220}{302.5} \simeq 0.73\ A$$

16-Solu Energy used by 250 watt T.V. set in 1 hour (3600 sec) is $E_1 = P_1 \times t_1 = 250 \times 3600 = 9 \times 10^5\ J$

Energy used by 1200 W toaster in 10 minutes (600 sec) is

$$E_2 = P_2 \times t_2 = 1200 \times 600 = 7.2 \times 10^5\ J$$

Therefore, energy used by TV is more.

17-Solu Here $R = 44\ \Omega$, $I = 5\ A$

Rate of heat developed $= P = I^2R = (5)^2 \times 44$

$= 1100\ W$